

API Documentation

Overview

Base URL: `http://localhost:5000` (development) | `https://api.example.com` (production)

Response Format: JSON

Authentication: Bearer token using Spring Security JWT

Rate Limits: 100 requests per minute

Authentication

Bearer Token

1. Obtain a token by logging in through the `/api/auth/login` endpoint with username and password.
2. Include in request headers: `Authorization: Bearer <token>`.
3. Token expires in 1 hour; refresh using the `/api/auth/refresh` endpoint.

Example Request with Auth

```
curl -X GET http://localhost:5000/api/leaves \  
-H "Authorization: Bearer your_token_here" \  
-H "Content-Type: application/json"
```

Endpoints

POST /api/leaves

Submit a new leave application.

Request:

- **Method:** POST
- **Path:** `/api/leaves`
- **Headers:** `Authorization: Bearer <token>`, `Content-Type: application/json`
- **Request Body:**

```
{  
  "employeeId": "UUID, unique identifier for employee",  
  "leaveType": "SICK, VACATION, PERSONAL",  
  "startDate": "YYYY-MM-DD, start date of the leave",  
  "endDate": "YYYY-MM-DD, end date of the leave"  
}
```

Response:

- **Status 201 (Created):**

```
{
  "id": "UUID, unique identifier for leave request",
  "status": "PENDING"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 500 (Server Error):** Unexpected error

Example:

```
curl -X POST http://localhost:5000/api/leaves \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json" \
-d '{"employeeId": "123e4567-e89b-12d3-a456-426614174000", "leaveType": "SICK", "startDate":
```

GET /api/leaves/{id}

Retrieve leave application details.

Request:

- **Method:** GET
- **Path:** /api/leaves/{id}
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json
- **Query Parameters:**
- **id (required):** UUID, unique identifier for leave request

Response:

- **Status 200 (OK):**

```
{
  "id": "UUID, unique identifier for leave request",
  "employeeId": "UUID, unique identifier for employee",
  "leaveType": "SICK, VACATION, PERSONAL",
  "startDate": "YYYY-MM-DD, start date of the leave",
  "endDate": "YYYY-MM-DD, end date of the leave",
  "status": "PENDING, APPROVED, REJECTED"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field

- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave application not found
- **Status 500 (Server Error):** Unexpected error

Example:

```
curl -X GET http://localhost:5000/api/leaves/123e4567-e89b-12d3-a456-426614174000 \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

PUT /api/leaves/{id}/approve

Approve a leave application.

Request:

- **Method:** PUT
- **Path:** /api/leaves/{id}/approve
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json
- **Query Parameters:**
- **id (required):** UUID, unique identifier for leave request

Response:

- **Status 200 (OK):**

```
{
  "id": "UUID, unique identifier for leave request",
  "status": "APPROVED"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave application not found
- **Status 500 (Server Error):** Unexpected error

Example:

```
curl -X PUT http://localhost:5000/api/leaves/123e4567-e89b-12d3-a456-426614174000/approve \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

PUT /api/leaves/{id}/reject

Reject a leave application.

Request:

- **Method:** PUT
- **Path:** /api/leaves/{id}/reject
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json
- **Query Parameters:**
- **id (required):** UUID, unique identifier for leave request

Response:

- **Status 200 (OK):**

```
{
  "id": "UUID, unique identifier for leave request",
  "status": "REJECTED"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Leave application not found
- **Status 500 (Server Error):** Unexpected error

Example:

```
curl -X PUT http://localhost:5000/api/leaves/123e4567-e89b-12d3-a456-426614174000/reject \
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

GET /api/leaves/balance/{employeeId}

Get employee leave balance.

Request:

- **Method:** GET
- **Path:** /api/leaves/balance/{employeeId}
- **Headers:** Authorization: Bearer <token>, Content-Type: application/json
- **Query Parameters:**
- **employeeId (required):** UUID, unique identifier for employee

Response:

- **Status 200 (OK):**

```
{
  "employeeId": "UUID, unique identifier for employee",
  "leaveType": "SICK, VACATION, PERSONAL",
  "balance": "integer, remaining leave balance"
}
```

- **Status 400 (Bad Request):** Missing or invalid required field
- **Status 401 (Unauthorized):** Invalid or expired token
- **Status 403 (Forbidden):** Insufficient permissions
- **Status 404 (Not Found):** Employee not found
- **Status 500 (Server Error):** Unexpected error

Example:

```
curl -X GET http://localhost:5000/api/leaves/balance/123e4567-e89b-12d3-a456-426614174000
-H "Authorization: Bearer token" \
-H "Content-Type: application/json"
```

Error Codes and Meanings

Status	Code	Meaning	When It Occurs
400	BAD_REQUEST	Invalid input	Submit leave application with invalid employee ID
401	UNAUTHORIZED	Invalid credentials	Token missing or expired
403	FORBIDDEN	Access denied	User lacks required role
404	NOT_FOUND	Resource not found	Leave application or employee not found
409	CONFLICT	Duplicate or state conflict	Record already exists
429	RATE_LIMIT	Too many requests	Exceeded rate limit
500	INTERNAL_ERROR	Server error	Unexpected error, check logs

Rate Limits and Quotas

- 100 requests per minute: Exceeding this limit will result in a 429 RATE_LIMIT response.

- 1000 requests per hour: Exceeding this limit will result in a temporary block of further requests.

Data Types and Formats

- **Timestamps:** ISO 8601 format (e.g., 2026-01-15T10:30:00Z)
- **Booleans:** true or false
- **IDs:** UUID v4 or numeric, as documented per entity
- **Enums:** Specific values listed in request/response examples

Pagination (if applicable)

This API does not support pagination.

SDK Examples

No SDKs are provided for this API.